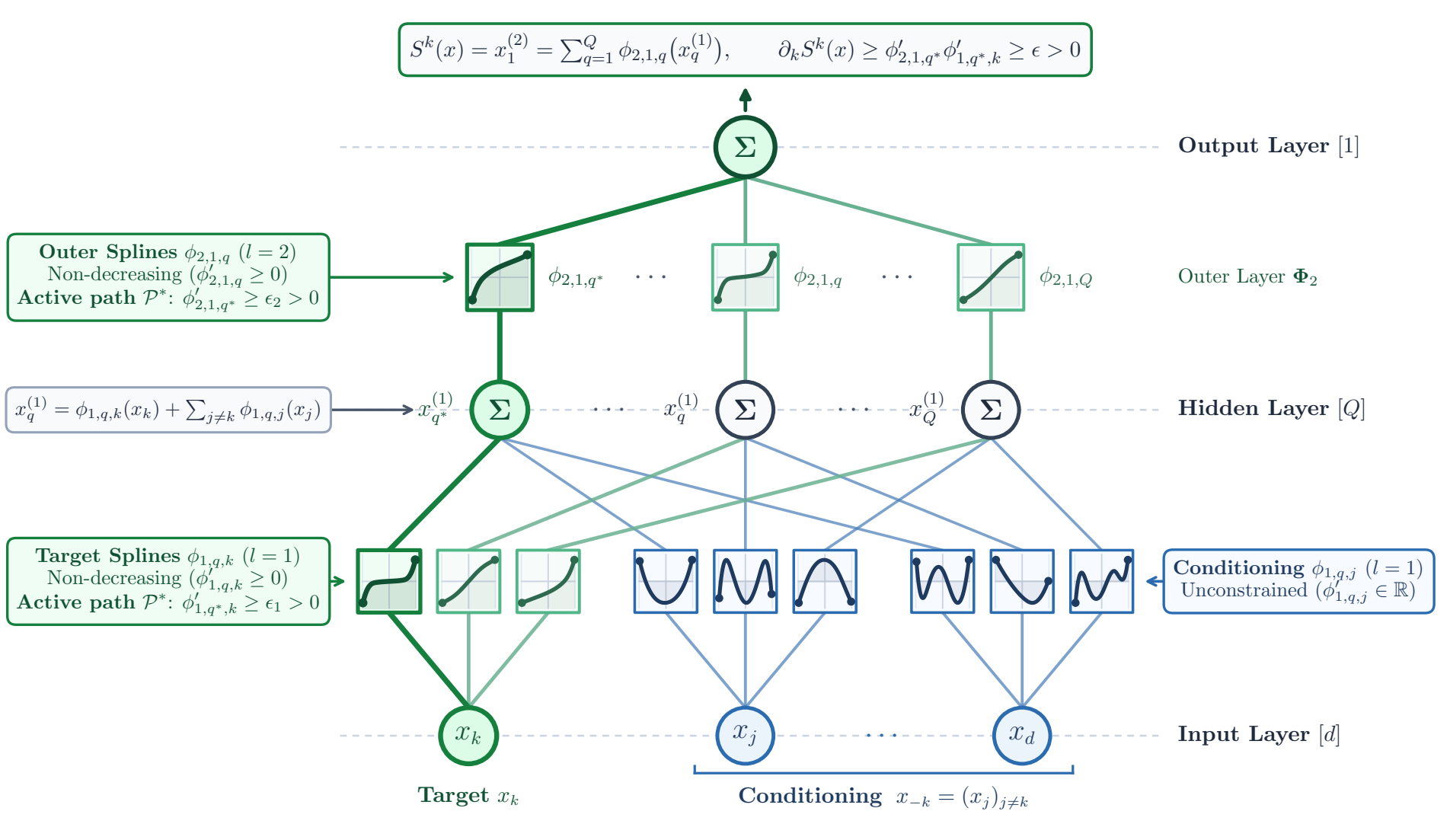

$S^k(x) = x_1^{(2)} = \sum_{q=1}^{Q} \phi_{2,1,q}\left(x_q^{(1)}\right), \qquad \partial_k S^k(x) \geq \phi'_{2,1,q^*}\phi'_{1,q^*,k} \geq \epsilon > 0$
Output Layer $[1]$
Outer Splines $\phi_{2,1,q}$ $(l=2)$
Non-decreasing $(\phi'_{2,1,q} \geq 0)$
Active path $\mathcal{P}^*$: $\phi'_{2,1,q^*} \geq \epsilon_2 > 0$
$\phi_{2,1,q^*}$
$\phi_{2,1,q}$
$\phi_{2,1,Q}$
Outer Layer $\mathbf{\Phi}_2$
$x_q^{(1)} = \phi_{1,q,k}(x_k) + \sum_{j \neq k} \phi_{1,q,j}(x_j)$
$x_{q^*}^{(1)}$
$x_q^{(1)}$
$x_Q^{(1)}$
Hidden Layer $[Q]$
Target Splines $\phi_{1,q,k}$ $(l=1)$
Non-decreasing $(\phi'_{1,q,k} \geq 0)$
Active path $\mathcal{P}^*$: $\phi'_{1,q^*,k} \geq \epsilon_1 > 0$
Conditioning $\phi_{1,q,j}$ $(l=1)$
Unconstrained $(\phi'_{1,q,j} \in \mathbb{R})$
x_k
x_j
x_d
Input Layer $[d]$
Target x_k
Conditioning $x_{-k} = (x_j)_{j \neq k}$